

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 1st Term Examination, 2017
CE 2107
(Construction Technology and Materials)

Full Marks: 210

Time: 3 hrs

- N.B.** i) Answer any **three** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable values for any missing data.

Section – A

1. (a) Define the following terms: (06)
(i) Available water holding capacity (ii) Cation-exchange capacity
(iii) Pedology
- (b) Write down the uses of mild steel, cast iron, wrought iron, aluminum and plastics. (15)
- (c) What is Building Stone? What are the common uses of Building Stones? (06)
- (d) What are meant by hack drying, natural bed of stones and dressing of stones? Why No. 4 sieve is named so? (08)
2. (a) What are the harmful constituents of brick clay? What are their harmful effects? (07)
- (b) What are the properties of good quality bricks? How can you identify good bricks in field? (10)
- (c) What is frog mark? Why is it used? (04)
- (d) What is the difference between "wood" and "timber"? Discuss different types of seasoning of timber with advantages. (14)
3. (a) Write down the differences between: (09)
(i) Initial setting time and final setting time of cement.
(ii) False setting and flash setting of cement.
(iii) Hydration and soundness of cement.
- (b) Classify sand depending on FM values, Which one is good for mortar and plastering? (05)
- (c) Describe the process of manufacturing portland cement by wet process with neat flow diagram. (10)
- (d) Grain size analysis data of a river sand and sea sand is given below. Calculate the combined Fineness modulus when these two sands are mixed with each other at equal amount. (11)

Sieve No.	Weight Retained (gm)	
	Sea Sand	River Sand
4	10	20
8	60	10
16	90	30
20	60	20
30	110	25
50	20	250
60	50	150
100	100	50
200	85	30
Pan	15	15

4. (a) What is concrete? Define "Bleeding" and "Segregation" of concrete. What are their effects on quality of concrete? Write down the properties of fresh concrete. (17)

- (b) What is meant by workability of concrete? Give a comparison between cube specimen and a cylinder specimen in concrete compressive strength test. (10)
- (c) What is curing of concrete? What are the factors on which properties of concrete depend? (08)

Section – B

5. (a) What is foundation? On what condition a deep foundation is provided? (04)
- (b) What is pile foundation? Classify pile according to their functions and material used. (09)
- (c) Write short notes with neat sketches on– (16)
 - i) Earthen cofferdam ii) Isolated column footing iii) Raft Foundation iv) Caisson
- (d) Draw a typical cross section of a framed structure. (06)

6. (a) What is load bearing and non-load bearing wall? Why bonds are important for masonry walls? (08)
- (b) What is Efflorescence? How it can be removed from the wall surface? (07)
- (c) List the governing factors for the selection of suitable type of floor. (08)
- (d) Write down the functions of components of floor. (04)
- (e) What is cavity wall? What are the main advantages of cavity wall? (08)

7. (a) What is road pavement? Write down the functions of a road pavement? (05)
- (b) What are the factors for the choice of a pavement type? (04)
- (c) What do you mean by the term "Rigid Pavement"? What are the components of rigid pavement? (05)
- (d) Write short notes on (09)
 - (i) Improved sub-grade (ii) Tie bars (iii) Flexible pavement
- (e) What is "Frost Heave" and "Spring Break-up"? Write down the measures that can protect frost action. (12)

8. (a) What are the factors that are considered at the time of modern building construction? (05)
- (b) How excavation is proceeded up to a large depth for the trench of a building? Explain with figure. (12)
- (c) Differentiate among superstructure, sub-structure and foundation. (09)
- (d) What are the aspects of using Composite structures? (09)

Khulna University of Engineering & Technology
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BURP 2nd Year 1st Term Examination, 2017

Hum 2117
(Sociology for Planners)

Full Marks: 210

Time: 3.00 hrs

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Define Sociology and Urban Sociology. (10)
(b) Explain the importance of studying Sociology for the students of Urban and Regional Planning (URP) in Bangladesh. (10)
(c) Explain the nature and character of different societies. (15)
2. (a) What is group? Explain different types of group and group size. (10)
(b) What is family? Describe the impacts of globalization in changing role of family in Bangladesh. (10)
(c) What is socialization? Explain that 'self' is the heart of the socialization. (15)
3. (a) Explain the taxonomy of culture. (10)
(b) What are the bases of human behavior? (10)
(c) Explain the common carriers of culture that share all over the world. (15)
4. (a) What is religion? Explain the elementary form of religion. (10)
(b) What is 'modernization'? Explain the changes in social structure in modernizing countries. (10)
(c) Explain the Demographic Transition theory of population with necessary example. (15)

Section – B

5. (a) What is urbanization? Describe evolution of city with characteristics of each phase. (10)
(b) Explain the relationship between industrial revolution and urbanization. (10)
(c) What are the major characteristics of Industrial city? (15)
6. (a) What is sub-urbanization? Explain causes and consequences of sub-urbanization. (15)
(b) What is Urbanism? Describe social characteristics of urbanism. (10)
(c) Explain 'Growth Machine Theory' in the light of rapid urbanization in Bangladesh. (10)
7. (a) What is urban ecology? (05)
(b) What is urban growth? As a student of URP which urban growth model do you prefer for Bangladesh? Why? (15)
(c) What is social control? Explain the role of social control in urban areas to prevent deviant behaviour. (15)
8. (a) What is migration? Explain Everett S Lee's Laws of migration. (15)
(b) What is environmental pollution? Describe causes and possible remedies of most pervasive environmental pollution. (10)
(c) How will population growth affect global ecology? (10)

Khulna University of Engineering & Technology
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BURP 2nd Year 1st Term Examination, 2017

URP 2113
(Site and Area Planning)

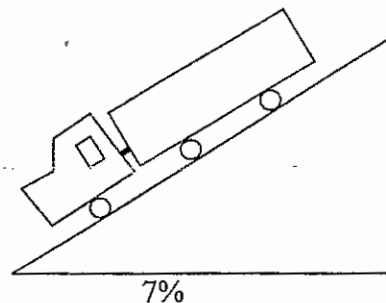
Full Marks: 210

Time: 3.00 hrs

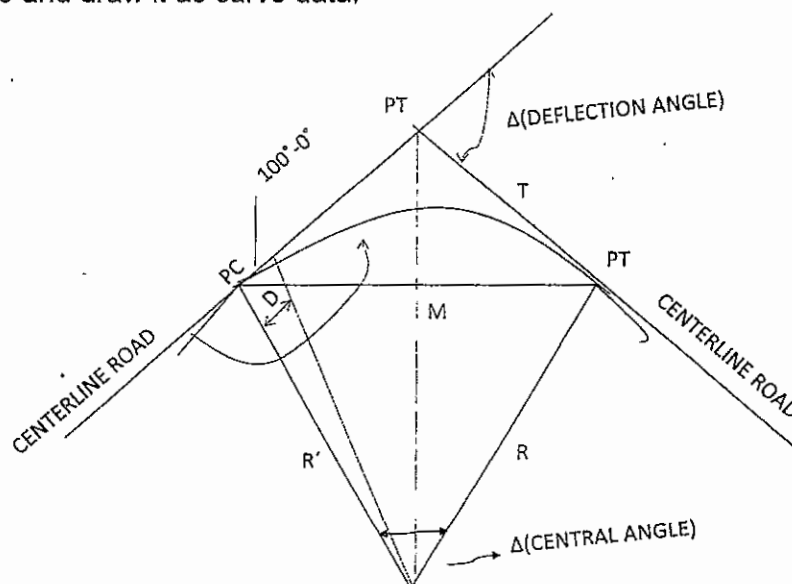
- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Define Site Planning according to Kevin Lynch and Gray Hack. (05)
(b) Kevin Lynch outlines an eight-stage Site Planning Cycle – Draw and explain the cycle. (10)
(c) "Site selection involves measuring the needs of a new project against the merits of potential locations" – Justify the statement with appropriate examples. (12)
(d) "Site Planning is involved with several professions viz. Urban Planning, Architecture and Civil Engineering" – briefly explain with your judgments. (08)
2. (a) Preparation of Site Inventory helps to do Site Analysis that includes various elements of a site. Give a short overview of the various elements of a site. Give a short overview of the various elements of a Site Inventory Checklist. (15)
(b) What is slope? What does the 7% mean in the following figure? (05)



- (c) Write down the Meteorological, Physical and Human factors affecting surface runoff. (15)
3. (a) How can land use controls achieve ecological sustainability? (15)
(b) Mention the jobs of a site planner needed for proper planning of a site. (05)
(c) In the following figure, the distance between the point of intersection from the beginning and end of the curve is 315 feet. Here, the central deflection angle is 60 degree. Assuming beginning at the curve at station 20+00, find the length of the curve and draw it as curve data. (15)



- 4 (a) Mention the factors influencing the selection of Horizontal Alignment. (10)
- (b) Briefly describe the necessity of planning for Parks, Recreation and Open Space while designing a community. (15)
- (c) Describe Relative Humidity with the help of an example. (05)
- (d) Why is it important to know the sun path diagram in Site Planning? (05)

Section – B

5. Write down the objectives, performance criteria and prescriptive measures with proper figure and explanation for any five of the following elements of residential subdivision planning. (35)
 - (a) Density
 - (b) Allotment layout
 - (c) Private open space
 - (d) Landscaping
 - (e) Road design
 - (f) Access way design
 - (g) Utility services.
6. (a) What do you understand by neighbourhood planning? State the scope of neighbourhood planning. (10)
- (b) "Site planning is related to spirit of the place, character of the place, nature of the project and behavioural studies" – Justify the statement with your rationalities by providing necessary examples, if it requires. (10)
- (c) Briefly explain the development process of site planning by drawing proper diagram to show the relationship among different process components. (10)
- (d) Mention any five good features of a planned neighbourhood. (05)
7. (a) Write down your understanding about the purposes of neighbourhood planning. (10)
- (b) "Site planning is involved with several different profession viz. majorly urban planning, architecture, landscape architecture and civil engineering" – Justify the statement by drawing proper diagram to show the relationships among different professionals. (10)
- (c) "The size of the neighbourhood unit is expressed as the size for the population and the extent of the geographical area" – Briefly describe the statement with proper examples. (10)
- (d) Briefly describe the available technologies for site and area planning. (05)
8. (a) What are the regulatory bases of Private Residential Housing Land Development Rules 2012? (05)
- (b) Write down the general conditions for the approval of a private residential housing project under "the Private Residential Housing Land Development Rules 2012." (15)
- (c) Write down the associated maps, designs, and places other than detailed layout plan while applying for the approval of private residential housing project. (15)

Khulna University of Engineering & Technology

Department of Urban & Regional Planning

BURP 2nd Year 1st Term Regular Examination, 2017.

URP 2191

(Statistics for Planners-I)

Full Marks: 210

Time: 3.0 hrs

N.B.: i) Answer any **three** questions from each section in separate script

ii) Figures in the right margin indicate marks

Section – A

1. (a) Explain the difference between Histogram and Frequency Bar Diagram. (08)

(b) The table is given below: (27)

Class interval	5-9	10-14	15-19	20-24	25-29	30-34
No. of item	7	8	9	5	10	4

(i) Draw frequency polygon in graph paper

(ii) Ogive

(iii) Calculate Mean, Median and Mode

(iv) Also locate mean, median and mode in the above frequency polygon.

2. (a) A sample of daily production was organized into the following distribution. (20)

Daily production	80-90	90-100	100-110	110-120	120-130
Frequency	13	7	15	20	8

Calculate (i) Variance and coefficient of variation;

(ii) Mean deviation about median (μ_e).

(b) Discuss briefly about the importance of statistics for urban planners. (06)

(c) The median and mode of the following wage distribution are Tk=33.50 and Tk=34 respectively. However, three frequencies are missing. Find their values. (09)

Wage (in thousands)	Frequency	Wage (in thousands)	Frequency
0-10	4	40-50	?
10-20	16	50-60	6
20-30	?	60-70	4
30-40	?	Total	230

3. (a) Define and explain the raw moments and central moments. (10)

(b) Calculate (i) Q_3 (3rd-quartile), (ii) D_7 (7th-decile), (iii) P_{67} (67th-percentile) from the following data: (15)

Class	Frequency	Class	Frequency
11-13	12	21-23	25
13-15	13	23-25	8
15-17	15	25-27	10
17-19	20	27-29	5
19-21	32		

(c) The workers' age are given below: 20, 22, 25, 30, 35, 40. Compute Kurtosis and also comment on the nature of the distribution. (10)

4. (a) Define Skewness and Kurtosis. Compute the coefficient of skewness from following data: (17)

10-20	20-30	30-40	40-50
5	7	9	10

(b) Define skew curve and normal curve. (06)

- (c) For two factories A and B belonging to the same industry, following details are available (12)

	Factory A	Factory B
No. of wage earners	100	80
Average wage monthly	Tk = 4600	Tk = 4900
S. D	Tk = 500	Tk = 400

- (i) Which factory between A and B pays larger amount as monthly wages?
(ii) Which factory shows greater variability in the distribution of wages?

Section – B

5. (a) Define with example (any three) (12)

(i) Sample space (ii) Conditional probability (iii) Probability mass function (iv) Elementary event.

- (b) Discuss the concept of probability of several methods including limitations (if any). (10)

- (c) The probability that husband, wife and child will be alive from now is given by 0.6, 0.5 and 0.9 respectively. Find the probability that within 30 years (13)

- (i) Both husband and wife will be alive
(ii) None of them will be alive (husband, wife and child)
(iii) At least one of them will die
(iv) At most one of them will die
(v) If in a city there are such 10,000 families, how many families do you expect in which none of them will die?

6. (a) Three components are packed in a box. Four events A, B, C and D are defined as follows: (18)

A = All three components are found to be in proper working order.
B = At least one component are found to be defective.
C = Exactly one component is found to be in proper working order.
D = None of the components is in proper working order.

- (i) Define above events by the elements of set.
(ii) Define the sample space of the three components of a box.
(iii) Interpret the following events: $A \cup B$, $B \cap C$, $A \cup B \cup C$.
(iv) Find $P(A)$, $P(B \cap C)$, $P(B \cap D)$.

- (b) Are the following function probability distribution functions, why? If yes, then find $E(x)$ of the function: (17)

(i)

x	1	-1	0
g(x)	2k	k	3k

(ii)

x	0	1	2	3
h(x)	0.5	0.4	0.3	-0.2

- (iii) $f(x) = x/100$; $-1 < x < 0$
 $= 0$; otherwise.

7. (a) Define Binomial distribution with necessary assumptions. Hence, find mean, third moment and coefficient of skewness of Binomial distribution. (22)

- (b) In a city there are 50,000 buildings. Each building contains 50 rooms. The probability of room contains nice air-flow is 0.001. Which distribution is good to find the following information, why? Hence find the expected number of building; (13)

- (i) At least one room contains nice air-flow.
(ii) All rooms contain nice air-flow.
(iii) Exactly two rooms contain nice air-flow.

8. (a) Define normal distribution and point out its five properties. (05)

- (b) In a city the average age is 40 yrs and standard deviation is 5 yrs. Find the probability [also indicate it in figure] of (12)

- (i) fertility people [i.e. $20 \leq \text{age} \leq 60$]
(ii) Infertility people [i.e. $0 \leq \text{age} \leq 20$ and $\text{age} > 60$]
(iii) If there are 50,000 people in that city then find expected number of child infertility people. [i.e. $\text{age} < 20$]

- (c) Fit the following data in Normal distribution given that $\mu = 35$ and $\sigma^2 = 4$. (18)

Age	10-20	20-30	30-40	40-50
Frequency	4	8	10	7

Also test the goodness of fit with 5% level of significance.

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 1st Term Regular Examination, 2017

URP2115
(Urban Planning Techniques)

Full Marks: 210

Time: 3.00hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Urban Development Plan ensures more equitable distribution of the benefits of development. - How? Explain with examples in context of Bangladesh. (15)
(b) Specify some components of Urban Development Plan with appropriate examples. (12)
(c) Draw a hierarchical order using different forms of urban development plan practiced in Bangladesh. Do you think the hierarchical order is suitable for the country? (08)
2. (a) "Master Plan is necessary to control the development in a systematic way". Explain the statement with appropriate example. (10)
(b) Briefly discuss the characteristics of Structure Plan and give some examples of strategic decisions in Structure Plan as that would affect the whole urban dynamics of Khulna City. (15)
(c) Make a list of "Key Policies" mentioned in DMDP Structure Plan. (10)
3. (a) "Strategic decisions and policy formulation are the logical outcome of national development goals in Strategic Plan". Explain this with examples. (10)
(b) Make a comparative analysis among Strategic Plan, Structure Plan and Detailed Area Plan with necessary examples in context of Bangladesh. (15)
(c) Briefly discuss about the forms of Local Plan with examples. Which form is mostly practiced in Bangladesh? (10)
4. (a) As a planner, in case of developing a new residential area in Bangladesh which technique will you follow between "Guided Land Development" and "Land Readjustment Plan"? -Why? (15)
(c) Make a comparative analysis among Area Improvement Plan, Infrastructure Led Development Plan and Land Ceiling techniques of participatory area planning. Which one is more feasible regarding socio-economic condition of Bangladesh? (12)
(c) Specify the implementation challenges of "Land Banking" technique in Bangladesh. (08)

Section – B

5. (a) Define Population "Estimate," "Forecast," & "Projection." Explain why (10)
and how Population Projections are often mislabeled as Forecast.
- (b) "Population extrapolation is a two-step process"- explain with the help (05)
of necessary diagram.
- (c) Articulate the difference between the Gompertz Curve and the Logistic (10)
Curve.
- (d) Following are the two curve expressions: (10)
Linear: $Y = a + bx$
Geometric: $Y = ab^x$
Which curve gives best possible fit to explain the population growth
trend of Bangladesh? Explain.

6. (a) What do you understand by Population Cohort Component Technique? (07)
Mention the types of Cohorts in practice and explain any one of them.
- (b) Explain the following terms with their associated equations (08)
i. Minimum Requirement Approach
ii. Shift Share Approach.
- (c) Write down the Problems of LQ Approach. (05)
- (d) Using LQ Approach, identify the following economic sectors as either (15)
basic or non-basic.

SIC code	Industry	Local	National
A	Agricultural	5240	70,96,533
C	Construction	1378	4642735
D	Food production	1687	16,014
I	Hotels and Lodgings	124	1,33,4210
H	Local Government	4060	9,32,6667
Total		12,489	22,41,6159

7. (a) Define the term "Affordable Housing." Illustrate the conceptual (10)
framework for Housing Choice.
- (b) The current amount of space used by commercial businesses and (15)
offices in 2017 is 8640000 sq. ft. and the population is 8640. Total
expected number of additional people by the year 2025 is 864, estimate
the need for new commercial space. Assume that the need for new
space per person will increase at the same rate throughout your
planning time period.
- (c) What is "Floor Area Ratio (FAR)"? If in an industrial area allowable site (10)
coverage is 60% and numbers of storey are fixed to 10, find out the
FAR.
8. (a) What is "urban renewal"? Diagrammatically represent the different (10)
levels of intervention for urban renewal.
- (b) What are the key questions to answer for land suitability analysis? (10)
Mention the basic steps of Land Suitability Analysis.
- (c) Briefly explain the three techniques for Parks, Recreation and Open (15)
Space Need Assessment.